Audio Amplifier Design and Implementation

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***Abstract*—This project aims to design and implement an audio amplifier with given specifications and performance criteria. The amplifier consists of four main stages: pre-amplification, gain stage, filtering, and power amplification. The pre-amplifier is designed using a differential common-emitter configuration to ensure high input impedance and minimize noise. The gain stage employs a common-emitter amplifier to achieve the required amplification. A band-pass filter is incorporated to limit unwanted frequency components while maintaining signal integrity. The power amplifier, implemented as a Class AB design, ensures efficient power delivery with minimal distortion. The project also includes an analysis of distortion and slew rate to evaluate signal quality and circuit performance. The total harmonic distortion (THD) is minimized using filtering techniques, while the slew rate is optimized to handle high-frequency signals effectively. The final design provides a robust and efficient audio amplification system suitable for various applications.**

***Index Terms*—Audio amplifier, pre-amplification, gain stage, power amplification, signal processing, distortion, slew rate.**

I. Introduction

Amplifiers are essential in modern audio systems, ensuring weak signals are amplified without significant distortion. An ideal audio amplifier should not only increase signal strength but also maintain clarity, efficiency, and minimal noise. This project explores the design and implementation of an audio amplifier tailored for low-power applications while meeting key performance criteria.

The amplifier design is structured into multiple stages, each playing a critical role in enhancing the input signal. The pre-amplifier boosts the weak audio signal while maintaining a high signal-to-noise ratio. The gain stage provides the necessary amplification, while a band-pass filter ensures that only the desired frequency range (20 Hz–20 kHz) is retained. The final stage, a Class AB power amplifier, delivers sufficient power to drive the output without introducing significant distortion.

II. Specifications

The audio amplifier is designed to meet the following technical specifications:

- Supply Voltage: 0–5V
- Input Signal Voltage: 10–40mV (peak-to-peak)
- Voltage Gain: 500 (achieved through pre-amplification and gain stages)
- Frequency Response: 20 Hz to 20 kHz (audible frequency range)
- Output Power: 1.5W
- Load Resistance: 10■
- Total Harmonic Distortion (THD): Minimal

III. Stages

There are four stages in Audio Amplifier they are

- Pre Amp stage
- Gain Stage
- Active Band Pass Filter
- Power Amplifier

1) Stability Factor in Audio Amplifiers:

The **stability factor** in audio amplifiers measures how stable the amplifier is against changes in parameters, particularly **temperature variations** and **component aging**. It is crucial because **unstable amplifiers** can lead to unwanted effects like **thermal runaway**, **oscillations**, or **distorted output**.

a) Definition:: The stability factor (S) quantifies how much the **collector current (I_C)** changes with **changes in current gain** (β). It is commonly defined as:

$$S = \frac{dI_C}{d\beta}$$

The ideal value of S is **1**, which indicates **no change** in collector current despite variations in β. Higher values indicate **less stability**.

2) Why Is Stability Important in Audio Amplifiers?:

1) **Temperature Variations:**
   - Amplifiers generate heat during operation, causing changes in **transistor parameters**, especially (β).
   - A high stability factor can cause **thermal runaway**, leading to **distorted audio output or damage** to the amplifier.
2) **Component Aging:**
   - As components age, their parameters may change, particularly **transistors** and **capacitors**.
   - An amplifier with a **low stability factor** is more robust against these changes.
3) **Feedback and Stability:**
   - Audio amplifiers often use **negative feedback** to **improve stability** and reduce distortion.
   - Proper feedback design reduces the **sensitivity of the output to variations** in transistor parameters.

IV. Pre-Amplifier Design

The pre-amplifier is the first stage in the audio amplifier system, responsible for boosting weak input signals while maintaining low noise. To ensure proper operation, the preamp has a high input impedance and a low output impedance. A common-emitter differential amplifier configuration is chosen to achieve high gain and noise reduction.

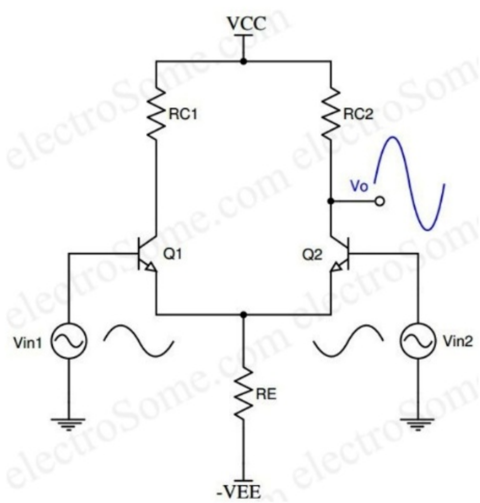


Fig. 1. Pre-Amplifier Circuit Diagram

V. Audio Amplifier Design and Performance

The audio amplifier designed in this project demonstrates excellent performance by achieving a high voltage gain, minimal distortion, and efficient power delivery. To further enhance signal quality, future improvements may include incorporating advanced filtering techniques and adaptive gain control.

VI. Role and Importance of a Pre-Amplifier

A well-designed pre-amplifier (preamp) is essential for maintaining signal clarity by minimizing interference and ensuring that the input signal is amplified without introducing unwanted modifications. Once the preamp boosts the signal, it is sent to the next stage for further amplification.

The primary purpose of the pre-amplifier stage is to provide initial amplification to the audio signal while minimizing noise intrusion. Ideally, the input resistance of the preamp should not be low, as this would cause the amplifier to draw high current, which the microphone cannot supply. This could lead to ineffective amplifier operation.

VII. Common-Emitter Differential Amplifier

The Common-Emitter Differential amplifier is particularly suitable for pre-amplification due to its high input and output impedance, coupled with good noise performance. Poor noise performance at the preamp stage can result in the already weak signal being completely masked by noise, compromising the audio quality. By effectively rejecting common-mode noise, the differential amplifier enhances signal integrity.

VIII. Differential Amplifier Applications

The differential amplifier is a fundamental building block in analog electronics, widely utilized in operational amplifiers and signal processing applications. Its primary function is to amplify the difference between two input signals while suppressing common-mode noise. This report includes an analysis of the differential amplifier's operation when one of its inputs, V_{in2}, is grounded.

The circuit consists of two identical NPN transistors Q_1 and Q_2 with a common emitter resistor R_E. The collectors are connected to the power supply VCC via resistors R_{C1} and R_{C2}, and the output voltage V_o is taken from the collector of Q_2. The input signals V_{in1} and V_{in2} are applied at the bases of the transistors, and the circuit operates with a negative power supply $-V_{EE}$ at the emitter resistor.

A. Biasing and DC Operation

When $V_{in2}=0$ (grounded), the base-emitter voltage of Q2 is determined as:

$V_E = V_{B2} - V_{BE2} = 0V - 0.7V = -0.7V$

The emitter current I_E is given by:

$I_E = |V_E - (-V_{EE})| / R_E$

Since I_E is shared between Q_1 and Q_2,

$I_{E1} + I_{E2} = I_E$

where I_{E1} and I_{E2} are the emitter currents of Q_1 and Q_2, respectively.

B. Effect of Input Voltage on Transistor Currents

When V_{in1} increases:

- The base-emitter voltage V_{BE1} of Q_1 increases, leading to an increase in I_{E1}.
- Since the total emitter current I_E remains nearly constant, an increase in I_{E1} results in a decrease in I_{E2}.
- Since collector current I_C is approximately equal to the emitter current in active mode, changes in emitter currents directly affect collector currents.

C. Output Voltage Behavior

The output voltage V_o is taken from the collector of Q_2 and is given by:

$V_o = VCC - I_{C2} R_C$

As I_{C2} decreases with an increasing V_{in1}, the voltage drop across R_C decreases, leading to an increase in V_o. This indicates that the circuit behaves as an **inverting amplifier**.

D. Voltage Gain of the Amplifier

The voltage gain A_v of the amplifier is determined by the transconductance g_m and collector resistance R_C:

$g_m = I_C / V_T$, $A_v = g_m R_C$

where V_T is the thermal voltage V_T = 25mV at room temperature.

E. Applications

When V_{in2} is grounded, the differential amplifier functions as a single-ended inverting amplifier. The output voltage increases when the input V_{in1} increases, and the voltage gain depends on the transconductance and the collector resistance. This configuration is widely used in:

- Operational amplifier input stages
- Signal amplification and processing
- Common-mode noise rejection
- Comparator circuits

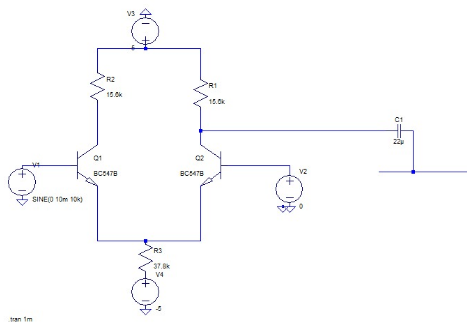


Fig. 2. LT Simulation 1st stage

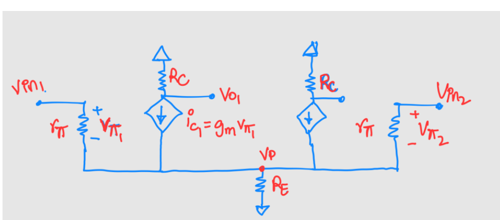


Fig. 3. small signal

Derivation

Applying KCL at V_p:

$$\frac{V_{\pi 1}}{r_\pi} + \frac{V_{\pi 2}}{r_\pi} + g_m V_{\pi 1} + g_m V_{\pi 2} = \frac{V_p}{R_E}$$

$$(V_{\pi 1} + V_{\pi 2}) g_m r_\pi + 1 \cdot \frac{R_E}{r_\pi} = V_p$$

$$(V_{in1} + V_{in2})(g_m r_\pi + 1)\frac{R_E}{r_\pi} = V_p \left[1 + 2 g_m r_\pi + 2\frac{R_E}{r_\pi}\right]$$

$$V_p = (V_{in1} + V_{in2}) \frac{(g_m r_\pi + 1)}{\frac{r_\pi}{R_E} + 2(g_m r_\pi + 1)}$$

Assuming $r_\pi \ll R_E$,

$$V_p = \frac{V_{in1} + V_{in2}}{2} \quad \rightarrow (1)$$

$$V_{\pi 1} = V_{in1} - V_p \quad \rightarrow (2)$$

$$V_{\pi 2} = V_{in2} - V_p \quad \rightarrow (3)$$

$$V_{o2} = \Theta g_m V_{\pi 2} R_C = -g_m (V_{in2} - V_p) R_C \quad \rightarrow (4)$$

Applying (1) in (4),

$$V_{o2} = -g_m (V_{in2} - V_{in1}) R_C \quad \rightarrow (5)$$

$$V_{o1} = -g_m \left(\frac{V_{in1} - V_{in2}}{2}\right) R_C \quad \rightarrow (6)$$

$$(V_{o2} - V_{o1}) = -g_m (V_{in2} - V_{in1}) R_C \quad \rightarrow (7)$$

$$\frac{V_{o2}}{(V_{in2} - V_{in1})} = -g_m R_C$$

$$\frac{V_{o2} - V_{o1}}{V_{in2} - V_{in1}} = -\frac{g_m R_C}{2}$$

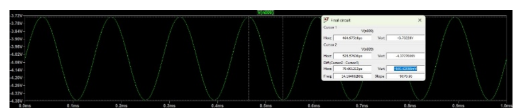


Fig. 4. LTSPICE Output

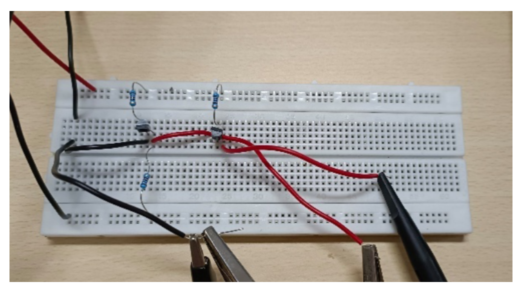


Fig. 5. Hardware 1st stage

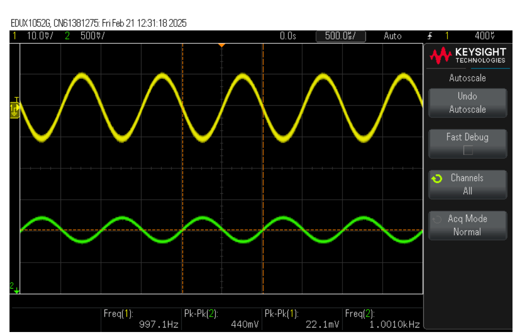


Fig. 6. Hardware output 1st stage

Common Mode Rejection Ratio

Common-Mode Rejection Ratio (CMRR) is a measure of a differential amplifier's ability to reject common-mode signals (noise) while amplifying differential signals. A higher CMRR indicates better noise rejection, making the amplifier more effective in reducing interference.

The mathematical expression for CMRR is given by:

$$\text{CMRR} = \frac{A_d}{A_{cm}}$$

where:

- A_d = Differential Gain (gain for the desired signal)
- A_{cm} = Common-Mode Gain (gain for noise or unwanted signals)

In decibels (dB), CMRR is expressed as:

$$\text{CMRR (dB)} = 20 \log_{10} \left(\frac{|A_d|}{|A_{cm}|} \right)$$

Example (CMRR = 10,000)

If CMRR is 10,000, it means the amplifier amplifies the differential signal 10,000 times more than the unwanted noise.

In decibels:

$$\text{CMRR} = 20 \log_{10}(10,000) = 80 \text{ dB}$$

This indicates that the amplifier significantly reduces noise and interference, ensuring clean signal amplification.

IX. Gain Stage

The gain stage is responsible for further amplifying the signal received from the pre-amplifier to achieve the required total gain of 500. This stage must be designed carefully to ensure signal integrity while minimizing distortion and instability.

A **common-emitter (CE) amplifier** is used due to its high voltage and current gain. It has a **low input impedance, allowing efficient signal reception, and a high output impedance**, making it suitable for driving the next stage.

To ensure stable operation, **biasing resistors** keep the transistor in its active region, preventing signal distortion. **Coupling capacitors** remove unwanted **DC components**, allowing only AC signals to pass. Additionally, proper impedance matching is necessary to ensure efficient signal transfer between stages, preventing signal loss and unwanted reflections that could degrade performance.

A. Basic Circuit Structure

The CE amplifier consists of the following essential components:

- A BJT transistor (NPN or PNP)
- Base resistor (R_B) for biasing
- Collector resistor (R_C) as load
- Emitter resistor (R_E) for stability (optional)
- Coupling capacitors at input and output
- Bypass capacitor across R_E (optional)

B. Biasing the Common Emitter Amplifier

Proper biasing ensures that the BJT operates in the active region, which is essential for linear amplification. The DC biasing establishes a quiescent operating point (Q-point) that allows for maximum signal swing without distortion. Common biasing methods include:

- Fixed bias configuration
- Voltage divider bias (most commonly used)
- Collector-to-base bias
- Emitter bias

X. Operating Principle

A. Signal Amplification Process

When an AC signal is applied to the base terminal of a properly biased CE amplifier, the following sequence of events occurs:

1) During the positive half-cycle of the input signal:
    - The forward bias across the emitter-base junction increases
    - This increases the flow of electrons from the emitter to the collector through the base
    - The collector current (I_C) increases
    - The increased I_C causes a larger voltage drop across the collector resistor R_C
    - The collector voltage (V_C) decreases
2) During the negative half-cycle of the input signal:
    - The forward bias across the emitter-base junction decreases
    - The flow of electrons from emitter to collector reduces
    - The collector current (I_C) decreases
    - The reduced I_C causes a smaller voltage drop across the collector resistor R_C
    - The collector voltage (V_C) increases

B. Mathematical Analysis

The voltage gain (A_v) of a CE amplifier is given by:

$$A_v = -\frac{R_C}{r_e + R'_E} \tag{1}$$

Where:

- R_C is the collector resistance
- r_e is the AC resistance of the emitter-base junction ($r_e = \frac{25mV}{I_E}$ at room temperature)
- R'_E is the unbypassed portion of the emitter resistance

The negative sign indicates phase inversion of the output with respect to the input.

For small-signal analysis, the h-parameter model gives:

$$A_v = -\frac{h_{fe} \cdot R_C}{h_{ie} + (1 + h_{fe})R'_E} \tag{2}$$

Where:

- h_{fe} is the forward current gain ()
- h_{ie} is the input impedance with output short-circuited

XI. Characteristics of Common Emitter Amplifier

A. Voltage Gain

The CE amplifier provides significant voltage gain, typically ranging from 10 to 300, depending on the circuit configuration and transistor parameters. This makes it ideal for applications requiring substantial voltage amplification.

B. Phase Shift

One distinctive characteristic of the CE amplifier is the 180-degree phase shift between input and output signals. This phase inversion must be considered when CE amplifiers are used in multi-stage configurations.

C. Input and Output Impedance

The CE amplifier exhibits moderate input impedance (typically 1-10 k) and relatively high output impedance (10-50 k). The input impedance is given by:

$$Z_{in} = R_B || (h_{ie} + (1 + h_{fe})R'_E) \tag{3}$$

And the output impedance is approximately:

$$Z_{out} \approx R_C \tag{4}$$

D. Frequency Response

The frequency response of a CE amplifier is limited by:

- Low-frequency cutoff due to coupling and bypass capacitors
- High-frequency cutoff due to junction capacitances and Miller effect

The miller capacitance significantly impacts the high-frequency response and is given by:

$$C_M = C_{CB}(1 + A_v) \tag{5}$$

Where C_{CB} is the collector-base junction capacitance.

XII. Applications

Common Emitter amplifiers find applications in various electronic systems:

- Audio pre-amplifiers and power amplifiers
- Radio frequency (RF) amplifiers
- Voltage amplification stages in communication systems
- Sensor interface circuits
- Instrumentation amplifiers

XIII. Practical Considerations

A. Thermal Stability

Temperature variations affect transistor parameters and can shift the Q-point, potentially causing distortion or circuit failure. Implementing negative feedback through an unbypassed emitter resistor improves thermal stability.

B. Signal Distortion

To minimize distortion and maintain signal integrity, the CE amplifier must be operated within its linear region. This requires:

- Proper biasing to establish an appropriate Q-point
- Limiting the input signal amplitude to prevent saturation or cutoff
- Using negative feedback to reduce nonlinearities

C. Bandwidth Enhancement Techniques

Several techniques can be employed to extend the bandwidth of CE amplifiers:

- Series inductive peaking
- Shunt capacitive peaking
- Negative feedback
- Cascode configuration

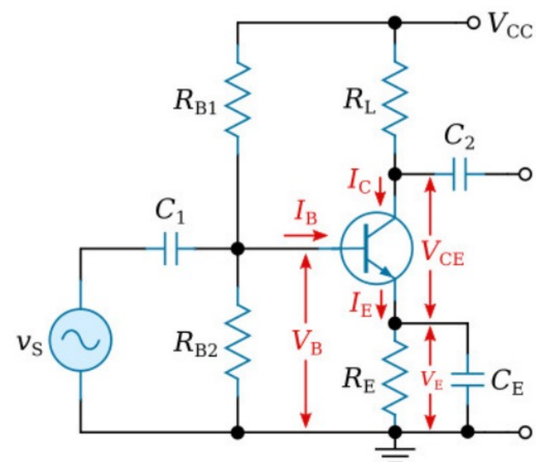


Fig. 7. CE-Amplifier

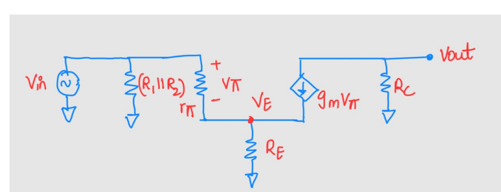


Fig. 8. Small signal simulation

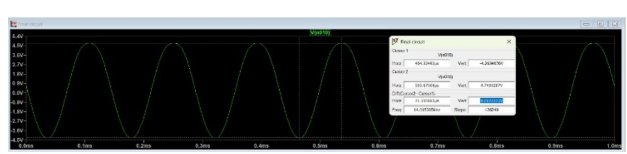


Fig. 9. LTSPICE Output

Gain Stage Derivation

Given:

$$V_{in} - V_E = V_\pi, \quad V_{out} = -g_m V_\pi R_C$$

Applying KCL at V_E:

$$\frac{V_\pi}{r_\pi} + g_m V_\pi = \frac{V_E}{R_E}$$

$$\frac{(V_{in} - V_E)}{r_\pi} + g_m (V_{in} - V_E) = \frac{V_E}{R_E}$$

$$V_{in}\left(\frac{g_m r_\pi + 1}{r_\pi}\right) = V_E\left(\frac{1}{r_\pi} + \frac{1}{R_E} + g_m\right)$$

Solving for V_E:

$$V_E = \frac{V_{in}(g_m r_\pi + 1)}{(g_m r_\pi + \frac{r_\pi}{R_E} + 1)}$$

Now, solving for V_{out}:

$$V_{out} = -g_m (V_{in} - V_E) R_C$$

$$= -g_m V_{in} \frac{\frac{r_\pi}{R_E} R_C}{(g_m r_\pi + \frac{r_\pi}{R_E} + 1)}$$

Since $g_m r_\pi = \beta \gg 1$,

$$V_{out} = -g_m V_{in} \frac{R_C}{(g_m + \frac{1}{R_E})}$$

Assuming $g_m \gg \frac{1}{R_E}$, we get:

$$\frac{V_{out}}{V_{in}} = \Theta \frac{R_C}{R_E}$$

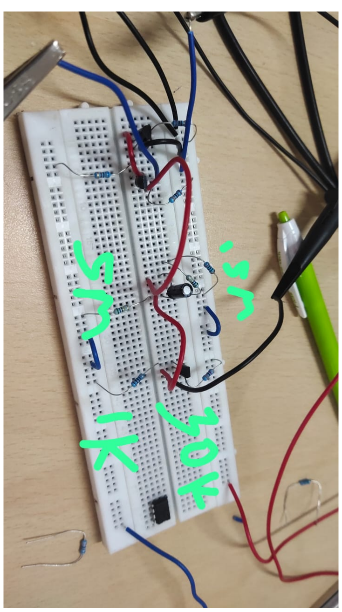


Fig. 10. Hardware Circuit till 2nd stage

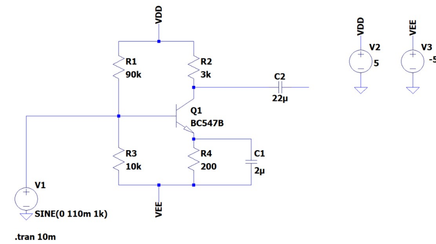


Fig. 11. Circuit diagram

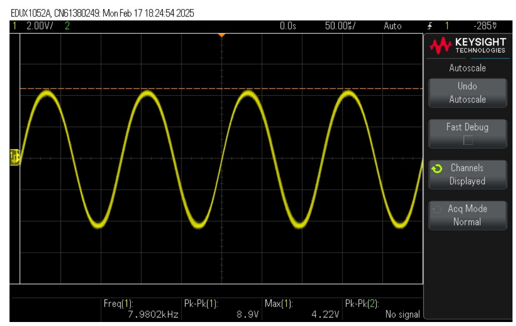


Fig. 12. Hardware Output

Key Components of CE Amplifier:

The CE amplifier consists of the following key components:

- Biasing Resistors (R_{B1}, R_{B2}):
- Provide proper DC biasing to set the transistor's operating point.
- Emitter Resistor (R_E):
- Stabilizes the operating point and improves linearity.
- Collector Resistor (R_L):
- Converts the amplified current into an output voltage.
- Coupling Capacitors (C_1, C_2):
- Block DC and allow AC signals to pass.
- Bypass Capacitor (C_E):
- Enhances gain by bypassing AC signal currents around RE.
- Power Supply (VCC):
- Provides the necessary operating voltage.

Operating Principle:

The CE amplifier operates in the active region of the transistor, ensuring amplification. The working mechanism is as follows:

1. **DC Biasing:**

   The resistors R_{B1} and R_{B2} form a voltage divider network to set the base voltage (VB). The emitter voltage is given by:
   $V_E = V_B - V_{BE}$

where V_{BE} 0.7V for a silicon transistor.

The collector-emitter voltage is:

$V_{CE} = V_{CC} - I_C R_L - I_E R_E$

2. **Signal Amplification:**

When an AC input signal (v_S) is applied, it modulates the base voltage (V_B). A small increase in V_B results in a larger increase in collector current (I_C) due to transistor current gain (). This increased I_C causes a larger voltage drop across R_L, leading to an inverted and amplified output voltage at the collector.

3. **Voltage Gain:**

The voltage gain (A_v) of the CE amplifier (without bypass capacitor C_E) is:
$A_v = -g_m R_L / (1 + g_m R_E)$
where g_m is the transconductance of the transistor, given by:
$g_m = I_C / V_T$, (where V_T 25mV at room temperature)
If C_E is present, the gain simplifies to:
Av -gmRL

4. **Phase Relationship**

The CE amplifier inverts the phase of the input signal, meaning that the output is 180° out of phase with the input.

5. **Frequency Response**

- *Low-Frequency Response:* Limited by coupling capacitors (C_1, C_2) and bypass capacitor (C_E).
- *Midband Response:* The amplifier operates with stable gain.
- *High-Frequency Response:* Limited by internal transistor capacitances.

Phase Inversion Happens in CE amplifier due to:

1. Base-Emitter Voltage (V_{BE}) controls Collector current (I_C)

- When the input signal at the base increases, the base-emitter voltage V_{BE} also increases.
- This increases the base current I_B, which in turn increases the collector current I_c (since I_c = SS I_B, where SS is the current gain of the transistor).

2. Collector voltage drops due to increased current.

- The collector resistor R_c is connected to Vcc, and the voltage at the collector is given by:

  $V_c = Vcc - I_c R_c$
- Since I_C increases, the voltage drop across R_C increases, which means the collector voltage Vc decreases.

3. Result: **Output is Inverted**

- If the input signal rises, the output voltage at the collector falls.
- If the input signal falls, the output voltage at the collector rises.
- This results in a 180° phase shift between input and output.

XIV. Active Band-Pass Filter

The filter stage is responsible for shaping the frequency response of the amplifier by allowing only the desired range of frequencies to pass while blocking unwanted signals. In this project, the filter ensures that the amplifier operates effectively within the audible frequency range of 20 Hz to 20 kHz without attenuating the input signal.

A band-pass filter is commonly used in audio applications to remove low-frequency noise (such as DC offsets) and high-frequency interference that could distort the output. This filter consists of a combination of capacitors and resistors, which define the cutoff frequencies and determine the overall response of the system.

To ensure stable and regulated power supply to various circuit components, a voltage regulation system is implemented. The filter stage operates with a ±12V power supply, which is then regulated to ±5V using a **voltage regulator**. This regulated voltage is supplied to other stages of the amplifier, ensuring consistent operation and preventing fluctuations that could impact performance.

A. Circuit Description

The given circuit is an active band-pass filter that uses:

- A Resistor-Capacitor (RC) network for both low-pass and high-pass filtering.
- An operational amplifier (op-amp) to provide gain and buffering.

B. Working Principle

1) **Input RC Network (High-Pass Filter):** The first RC network at the input acts as a high pass filter, blocking low-frequency components below its cutoff frequency (f_L) and allowing higher frequencies to pass.
2) **Feedback RC Network (Low-Pass Filter):** The second RC network in the feedback loop acts as a low-pass filter, allowing frequencies below its cutoff frequency (f_H) to pass while attenuating higher frequencies.
3) **Band-Pass Filtering:** The combination of the two RC networks results in a band-pass filter, allowing frequencies within the band ($f_L < f < f_H$) to pass while attenuating frequencies outside this range.

C. Circuit Analysis

The non-inverting terminal of the operational amplifier is connected to ground. Due to virtual shorting, the inverting terminal is also virtually grounded.

$$V_{in} - 0 = I \cdot Z_1 \tag{6}$$

$$0 - V_{out} = I \cdot Z_2 \tag{7}$$

$$\text{Gain} = -\frac{Z_2}{Z_1} \tag{8}$$

Where:

$$Z_1 = \frac{sR_1C_1 + 1}{sC_1} \tag{9}$$

$$Z_2 = \frac{R_2}{sR_2C_2 + 1} \tag{10}$$

D. Transfer Function

The transfer function $H(s)$, describing the frequency-dependent behavior, is given by:

$$H(s) = \frac{V_{out}}{V_{in}} = \frac{sR_2C_1}{(sR_1C_1 + 1)(sR_2C_2 + 1)} \tag{11}$$

where:

- $s = j\omega$ (Laplace variable)
- R and C are the resistor and capacitor values.

E. Key Parameters

1) Center Frequency (f_0):

$$f_0 = \frac{1}{2\pi\sqrt{R_1R_2C_1C_2}} \tag{12}$$

This is the frequency where the circuit achieves maximum gain.

2) Bandwidth (B):

$$BW = f_H - f_L \tag{13}$$

The range of frequencies passed by the filter.

3) Quality Factor (Q):

$$Q = \frac{f_0}{B} \tag{14}$$

A higher Q indicates a narrower pass band.

4) Cutoff Frequencies:

- Low cutoff frequency (f_L):

$$f_L = \frac{1}{2\pi R_1C_1} \tag{15}$$

- High cutoff frequency (f_H):

$$f_H = \frac{1}{2\pi R_2C_2} \tag{16}$$

F. Frequency Response

The frequency response of the circuit exhibits:

- **Attenuation:** For $f < f_L$ (low frequencies) and $f > f_H$ (high frequencies).
- **Passband:** Between f_L and f_H, centered at f_0.
- A peak gain at f_0, determined by the op-amp and feedback network configuration.

XV. Design Considerations

When designing this active band-pass filter, several factors should be considered:

A. Component Selection

- **Resistor Values:** Typical values range from 1 kto 100 k. Lower values consume more power but are less susceptible to noise.
- **Capacitor Values:** Typically range from 1 nF to 1 F. Consider using film capacitors for better tolerance and stability.
- **Op-amp Selection:** Choose an op-amp with:
  - Sufficient gain-bandwidth product (GBP) for the desired frequency range
  - Low input offset voltage and bias current
  - Adequate slew rate for the expected signal levels

B. Practical Implementation

- Use a dual power supply for the op-amp to handle bipolar signals
- Consider adding decoupling capacitors near the op-amp power pins
- Use a ground plane to minimize noise and interference
- Consider component tolerances when calculating the expected frequency response

XVI. Derivation

We choose

$$Z_1 = R_1 + \frac{1}{j\omega C_1}$$

$$Z_2 = R_2 \parallel \frac{1}{j\omega C_2} = \frac{\frac{R_2}{j\omega C_2}}{\frac{1}{j\omega C_2} + R_2} = \frac{R_2}{1 + j\omega R_2C_2}$$

$$\frac{V_{\text{out}}}{V_{\text{in}}} = -\frac{R_2 \cdot j\omega C_1}{(1 + j\omega R_2C_2)(1 + j\omega R_1C_1)}$$

$$\left|\frac{V_{\text{out}}}{V_{\text{in}}}\right| = \frac{\omega R_2C_1}{\sqrt{1 + (\omega R_2C_2)^2}\sqrt{1 + (\omega R_1C_1)^2}}$$

For

$$\omega \ll \frac{1}{R_1C_1} \ll \frac{1}{R_2C_2}$$

$$\left|\frac{V_{\text{out}}}{V_{\text{in}}}\right| = \omega R_2C_1 \implies 20\log\left|\frac{V_{\text{out}}}{V_{\text{in}}}\right| = 20\log(\omega R_2C_1)$$

For

$$\frac{1}{R_1C_1} \ll \omega \ll \frac{1}{R_2C_2}$$

$$\left|\frac{V_{\text{out}}}{V_{\text{in}}}\right| = \frac{R_2}{R_1} \implies 20\log\left|\frac{V_{\text{out}}}{V_{\text{in}}}\right| = 20\log\frac{R_2}{R_1}$$

For

$$\frac{1}{R_1C_1} \ll \frac{1}{R_2C_2} \ll \omega$$

$$\left|\frac{V_{\text{out}}}{V_{\text{in}}}\right| = \frac{1}{\omega R_1C_2} \implies 20\log\left|\frac{V_{\text{out}}}{V_{\text{in}}}\right| = -20\log(\omega R_1C_2)$$

XVII. Simulation and Testing

To verify the design, the following steps are recommended:

- Perform AC analysis simulation to verify the frequency response
- Measure the actual component values and recalculate the expected cutoff frequencies
- Test the circuit with various input frequencies to confirm the band-pass characteristics
- Measure the actual center frequency and bandwidth to compare with theoretical values

XVIII. Component Values and Calculations

For a band-pass filter with $f_L = 100$ Hz and $f_H = 10$ kHz:

Component	Value	Calculation
R_1	15.9 k	$R_1 = \frac{1}{2\pi f_L C_1}$
C_1	100 nF	(Selected)
R_2	1.59 k	$R_2 = \frac{1}{2\pi f_H C_2}$
C_2	10 nF	(Selected)
f_0	1 kHz	$f_0 = \frac{1}{2\pi\sqrt{R_1 R_2 C_1 C_2}}$
Q	0.1	$Q = \frac{f_0}{f_H - f_L}$

TABLE I
Component Values and Calculated Parameters

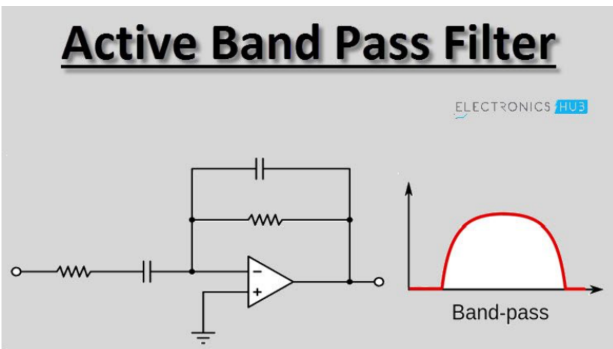


Fig. 13. Band pass filter

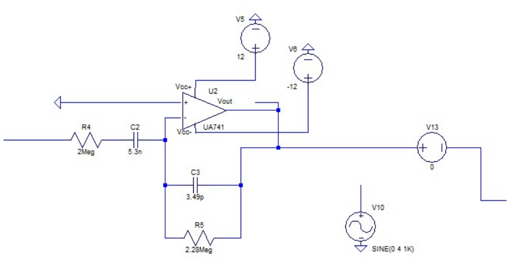


Fig. 14. LT spice simulation

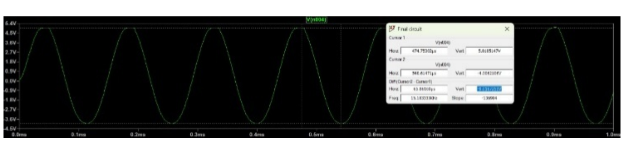


Fig. 15. LTSPICE Output

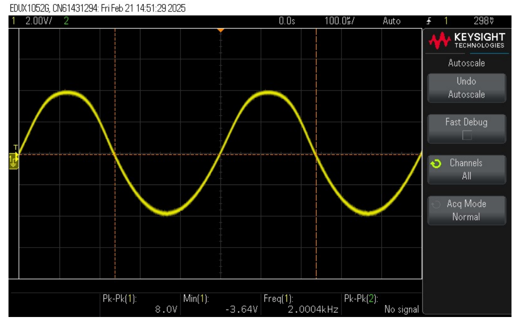


Fig. 16. 3rd stage output

Fig. 17. Frequency Response at 20 Hz

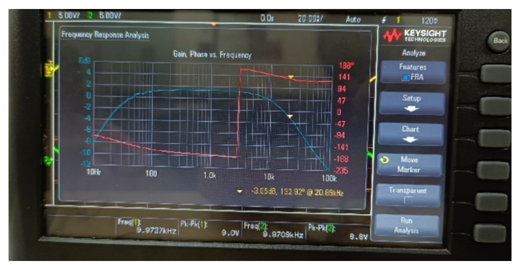


Fig. 18. Frequency Response at 20000 Hz

XIX. Power Amplifier

The power amplifier is the final stage of the audio amplification system, responsible for delivering sufficient power to drive the output load without introducing distortion. Unlike previous stages that focus on voltage amplification, the power amplifier is designed to provide the necessary current and power gain while maintaining signal integrity.

In this project, a Class AB power amplifier is used due to its efficiency and low distortion characteristics. Class AB amplifiers combine the advantages of Class A (low distortion) and Class B (high efficiency) amplifiers, ensuring better performance for audio applications. The design ensures that the amplifier does not introduce

XX. Power Amplifiers

Power amplifiers are used to increase the power level of a signal and are commonly classified into Class-A, Class-B, and Class-AB based on their operating point, conduction angle, and efficiency.

A. *Class-A Power Amplifier*

- **Operating Principle:** The transistor in a Class-A amplifier is biased so that it remains active (in the linear region) throughout the entire input signal cycle (360° conduction angle).
- **Features:**
  - High linearity, making it suitable for low-distortion amplification.
  - Poor efficiency, typically around $25\% - 30\%$.
  - High heat dissipation due to continuous current flow even without input signal.

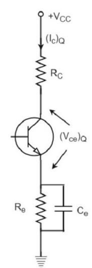


Fig. 19. Class A

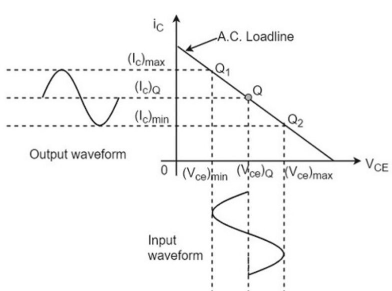


Fig. 20. Power Amp Characteristics

B. *Class-B Power Amplifier*

- **Operating Principle:** The transistor is biased at cutoff, and each transistor in a push-pull configuration conducts for half of the input signal cycle (180° conduction angle).
- **Features:**
  - Higher efficiency compared to Class-A, typically around $50\% - 70\%$.
  - Introduces crossover distortion due to the transition between transistors.
  - Requires a complementary pair of transistors (NPN and PNP or MOSFETs).

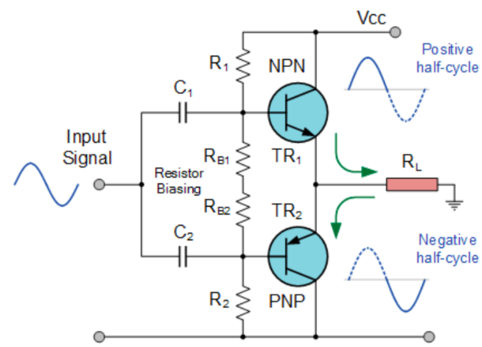


Fig. 21. Class B

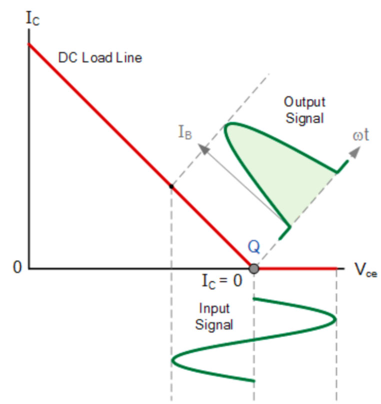


Fig. 22. Class B Output Characteristics Curves

Crossover distortion in a Class B amplifier occurs due to the **dead zone** between the **conduction of the NPN and PNP transistors**. This happens because:

1) **Diode Drop (0.6 - 0.7 V):**
   - Both transistors require a **base-emitter voltage (V_{BE})** of about **0.6 to 0.7 V** to start conducting.
   - Around the **zero-crossing point** of the input waveform, the input voltage is **not sufficient** to turn on either transistor.
   - As a result, **both transistors remain off**, leading to a **discontinuity in the output**.
2) **Non-Ideal Switching:**
   - The transition from the **NPN transistor to the PNP transistor** (or vice versa) is not instantaneous.
   - This transition causes a **small gap** where **neither transistor conducts**, resulting in **distortion at the zero-crossing point**.

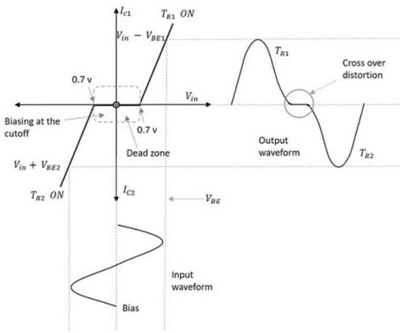


Fig. 23. Reasoning for COD

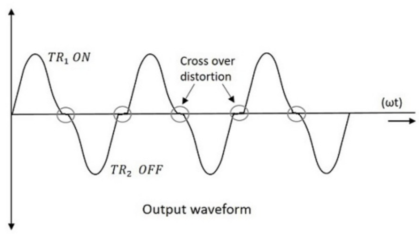


Fig. 24. Cross Over Distortion in Class B

C. Class-C Power Amplifier

- **Operating Principle:** Biased well below cutoff, conducting for less than 180° of the input cycle (typically 90° to 150°).
- **Features:**
  - Highest efficiency among conventional amplifier classes (up to 85%).
  - Significant nonlinearity and distortion.
  - Primarily used in RF applications where resonant circuits can filter out harmonics.
  - Not suitable for audio applications due to high distortion.

D. Class-D Power Amplifier

- **Operating Principle:** Uses pulse width modulation (PWM) to convert the input signal into a series of pulses, driving transistors as switches that are either fully on or fully off.
- **Features:**
  - Very high efficiency (up to 95%).
  - Reduced heat dissipation.
  - Requires output filtering to recover the original waveform.
  - Increasingly popular in audio applications due to improved designs.

E. Class-AB Power Amplifier

- **Operating Principle:** Combines the features of Class-A and Class-B by biasing the transistors slightly above cutoff, allowing both transistors to conduct for slightly more than half of the signal cycle (greater than 180° but less than 360° conduction angle).
- **Features:**
  - Improved linearity compared to Class-B, reducing crossover distortion.
  - Higher efficiency than Class-A, typically around 50%.
  - Moderate heat dissipation, balancing efficiency and linearity.

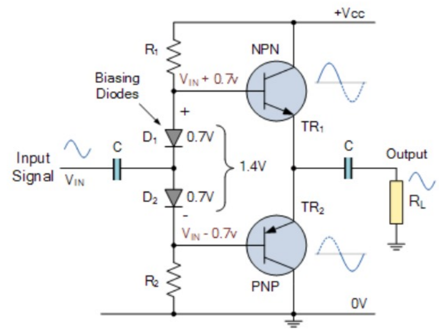


Fig. 25. Power Amplifier

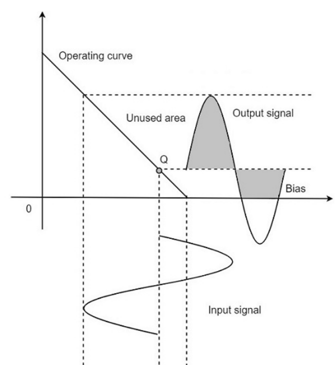


Fig. 26. Class AB Output Characteristics Curve

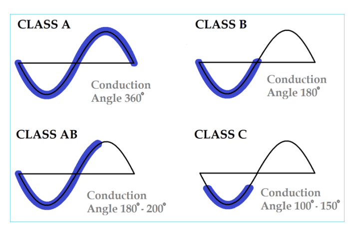


Fig. 27. All classes Area of conduction

Working of Power Amplifier:

A small constant current flows through the series circuit of R1-D1-D2-R2, producing voltage drops which are symmetrical either side of the input. With no input signal voltage applied, the point between the two diodes is zero volts. As current flows through the chain, there is a forward bias voltage drop of approximately 0.7V across the diodes which is applied to the base-emitter junctions of the switching transistors.

Therefore, the voltage drop across the diodes, biases the base of transistor TR1 to about 0.7 volts, and the base of transistor TR2 to about –0.7 volts. Thus, the two silicon diodes provide a constant voltage drop of approximately 1.4 volts between the two bases biasing them above cut-off.

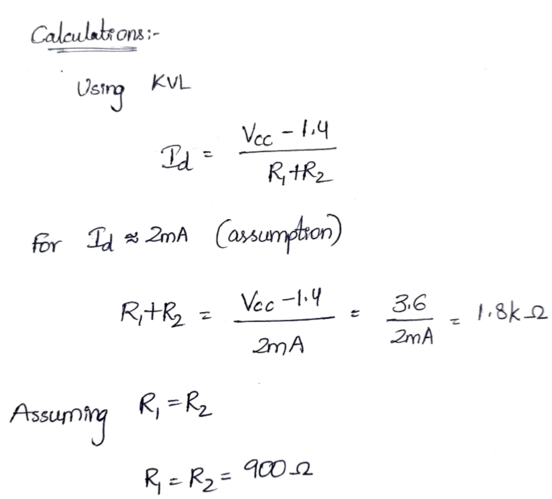
As the temperature of the circuit rises, so too does that of the diodes as they are located next to the transistors. The voltage across the PN junction of the diode thus decreases diverting some of the transistors base current stabilizing the transistors collector current.

If the electrical characteristics of the diodes are closely matched to that of the transistors base-emitter junction, the current flowing in the diodes and the current in the transistors will be the same creating what is called a current mirror. The effect of this current mirror compensates for variations in temperature producing the required Class AB operation thereby eliminating any crossover distortion.

In practice, diode biasing is easily accomplished in modern day integrated circuit amplifiers as both the diode and switching transistor are fabricated onto the same chip, such as in the popular LM386 audio power amplifier IC. This means that they both have identical characteristics curves over a wide temperature change providing thermal stabilization of the quiescent current.

The biasing of a Class AB amplifier output stage is generally adjusted to suit a particular amplifier application. The amplifiers quiescent current is adjusted to zero to minimize power consumption, as in Class B operation, or adjusted for a very small quiescent current to flow that minimizes crossover distortion producing a true Class AB amplifier operation.

In the above Class AB biasing examples, the input signal is coupled directly to the switching transistors bases by using capacitors. But we can improve the output stage of a Class AB amplifier a little more by the addition of a simple common-emitter driver stage as shown.



Calculations:-

Using KVL

$$I_d = \frac{V_{cc} - 1.4}{R_1 + R_2}$$

For $I_d \approx 2mA$ (assumption)

$$R_1 + R_2 = \frac{V_{cc} - 1.4}{2mA} = \frac{3.6}{2mA} = 1.8k\Omega$$

Assuming $R_1 = R_2$

$$R_1 = R_2 = 900\Omega$$

Fig. 28. Power Amplifier

We chose 2mA Bias current because:

1. Reduced Crossover Distortion:

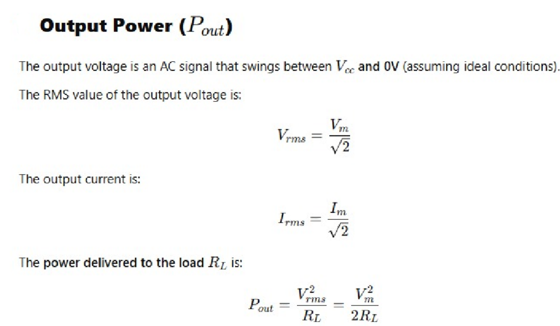
In a Class-AB amplifier, we slightly bias the transistors into conduction to avoid the dead zone where both transistors are off. A higher bias current (like 2mA) ensures that the transistors turn on smoothly, reducing crossover distortion.

2. Ensures Proper Operation of the Diodes:

Diodes require a small current to stay in conduction and maintain their 0.7V drop. At very low currents (<1mA), diodes may not operate consistently, leading to variations in base-emitter voltages (V $_{be}$).

3. Faster response and Better Linearity:

A slightly higher bias current allows the transistors to switch on/off faster, improving the amplifier's frequency response. It helps achieve better linearity in signal amplification.



Output Power (P_{out})

The output voltage is an AC signal that swings between V_{cc} and **0V** (assuming ideal conditions).

The RMS value of the output voltage is:

$$V_{rms} = \frac{V_m}{\sqrt{2}}$$

The output current is:

$$I_{rms} = \frac{I_m}{\sqrt{2}}$$

The **power delivered to the load** R_L is:

$$P_{out} = \frac{V_{rms}^2}{R_L} = \frac{V_m^2}{2R_L}$$

Fig. 29. Power Amplifier

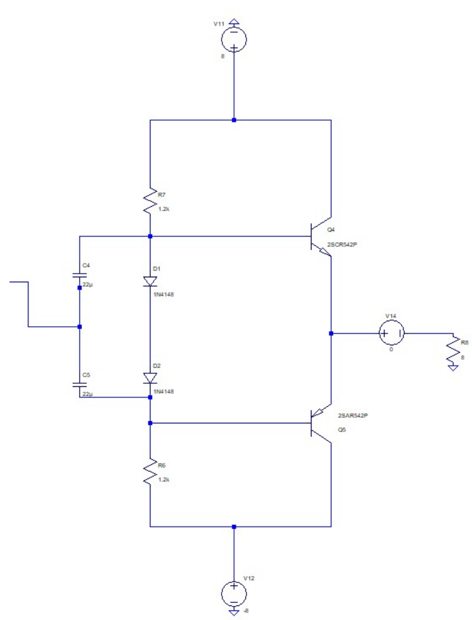


Fig. 30. LT spice simulation Power Amplifier

PRACTICAL CONSIDERATIONS

When selecting a power amplifier class, designers must consider:

- Trade-offs between efficiency, linearity, and distortion
- Power requirements and thermal management
- Frequency range of operation
- Load impedance matching for maximum power transfer
- Cost and complexity constraints

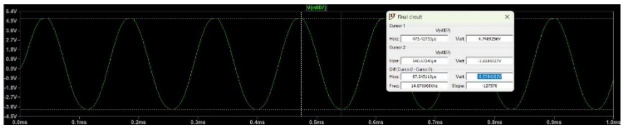


Fig. 31. Circuit Diagram

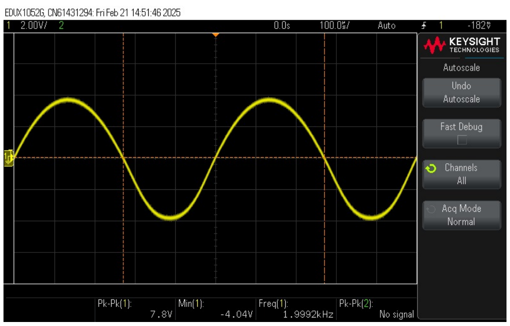


Fig. 32. 4th stage final output

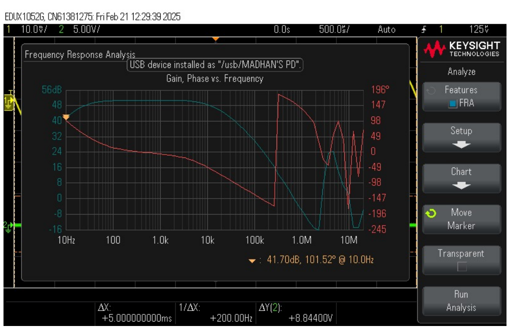


Fig. 33. Final Bode Plot

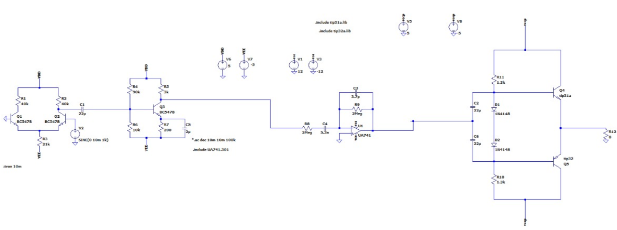


Fig. 34. Complete full circuit

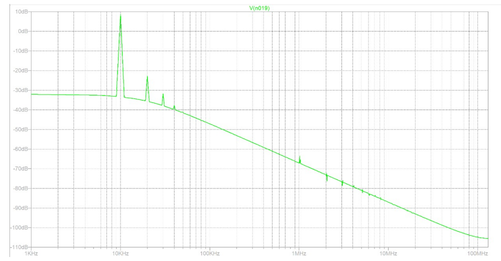


Fig. 35. Fast Fourier Transform

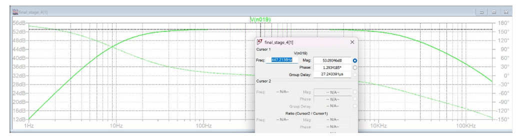


Fig. 36. Bode plot in simulation

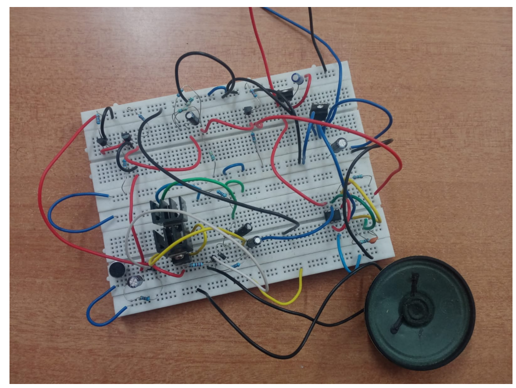


Fig. 37. Hardware complete Circuit Diagram

Now we will see about the Slew rate and its importance in each stage

XXI. WHAT IS SLEW RATE?

Slew Rate (SR) is the maximum rate of change of output voltage per unit time in an electronic circuit, usually measured in volts per microsecond (V/µs). It represents how fast the output of a circuit (such as an op-amp, amplifier, or ADC) can respond to a rapidly changing input signal.

Mathematically, it is given by:

$$S.R. = \max\left(\frac{dV_{\text{out}}}{dt}\right) \tag{17}$$

Where:

- dV_{out} = Change in output voltage
- dt = Change in time

XXII. WHY IS SLEW RATE IMPORTANT?

- **Prevents Distortion** – If the slew rate is not high enough, the circuit cannot keep up with fast-changing signals, leading to distortion.

- **Essential for High-Frequency Applications** – In audio, communication, and high-speed electronics, a higher slew rate ensures that signals are accurately reproduced.
- **Limits Circuit Performance** – Even if a circuit has a high bandwidth, a low slew rate can degrade performance, making it unsuitable for high-speed signals.

XXIII. Ideal vs. Practical Slew Rate

- **Ideally:** Slew rate should be infinitely high to respond instantly to any input change.
- **Practically:** Every circuit has limitations due to internal capacitances, resistances, and power supply constraints.

XXIV. How to Increase Slew Rate?

- **Increase Maximum Operating Voltage** – A higher voltage allows the circuit to drive the output faster.
- **Reduce Capacitive Impedance** – Internal capacitance slows down voltage changes, so minimizing capacitance helps improve the slew rate.
- **Use Faster Components** – High-speed op-amps and transistors designed for fast switching have better slew rates.

XXV. Example Calculation

For an amplifier with:

- Maximum output voltage swing $V_m = 15mV$
- Frequency f (varies based on input audio)

The required minimum slew rate is:

$$S = 2\pi f V_m \tag{18}$$

This means that for higher frequencies, the required slew rate increases to avoid distortion.

a) Preamp Stage (Preamplifier)::

- **Purpose:** Boosts low-level audio signals (e.g., from a microphone or instrument) to line level before sending them to the power amplifier.
- **Slew Rate Impact:**
  - The preamp stage needs a high slew rate to accurately amplify **high-frequency signals** without introducing distortion.
  - If the slew rate is too low, **high-frequency audio components** (like treble) can be **distorted**, resulting in **muddied or clipped sound**.
- **Application:** High-quality audio equipment and professional sound systems.

b) 2. Amplify Stage (Intermediate Amplifier Stages)::

- **Purpose:** Further amplifies signals after pre-amplification.
- **Slew Rate Impact:** Intermediate stages also require a sufficient slew rate to **maintain signal integrity** and avoid distortion before sending to the power amp.
- **Application:** Multi-stage amplification circuits in audio processing systems.

c) 3. Bandpass Filter Stage::

- **Purpose:** Allows signals within a certain frequency range to pass while attenuating others.
- **Slew Rate Impact:**
  - **Indirect relation:** If the bandpass filter uses **active components (like op-amps)**, the slew rate of those op-amps can affect the **sharpness and accuracy** of the filtered output.
  - A **low slew rate** in the amplifier used within the filter can **round off** sharp signal transitions, degrading the **filter performance**.
- **Application:** Equalizers and frequency-selective amplifiers.

d) 4. Power Amplifier Stage::

- **Purpose:** Boosts the audio signal to a level sufficient to drive speakers or other loads.
- **Slew Rate Impact:**
  - Power amplifiers require a high slew rate to **faithfully reproduce fast transients and high-frequency signals**.
  - Low slew rate causes **slew-induced distortion**, resulting in **harsh, distorted treble** or **muffled high-frequency sounds**.
- **Application: High-fidelity audio systems**, **home theater amplifiers**, and **professional sound reinforcement systems**.

1) Relation between stability factor and Slew Rate::

Although **stability factor and slew rate** address different aspects, they are indirectly related:

- **High stability** ensures consistent performance across temperature variations.
- **High slew rate** ensures the amplifier accurately follows rapid changes in the input.
- An unstable amplifier may experience **oscillations or clipping**, which affects both **signal fidelity** and **slew rate performance**.

Total Harmonic Distortion (THD)

Total Harmonic Distortion (THD) is a measure of harmonic distortion present in a signal. It is given by the formula:

$$\text{THD} = \frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \cdots + V_n^2}}{V_1}$$

Where:

- V_1 is the RMS voltage of the fundamental frequency.
- $V_2, V_3, V_4, \ldots, V_n$ are the RMS voltages of harmonic frequencies.

Percentage THD

THD is often expressed as a percentage:

$$\text{THD } (\%) = \left(\frac{\sqrt{V_2^2 + V_3^2 + V_4^2 + \cdots + V_n^2}}{V_1} \right) \times 100$$

THD Calculation

Given data:

$$f_1 = 8.62 \text{ dB}$$

$$V_{f_1,\text{RMS}} = \frac{2.699}{\sqrt{2}}$$

$$f_2 = -22.67 \text{ dB}$$

$$V_{f_2,\text{RMS}} = \frac{0.735}{\sqrt{2}}$$

$$f_3 = -31.78 \text{ dB}$$

$$V_{f_3,\text{RMS}} = \frac{0.025}{\sqrt{2}}$$

$$f_4 = -37.72 \text{ dB}$$

$$V_{f_4,\text{RMS}} = \frac{0.013}{\sqrt{2}}$$

The root sum of squares of the harmonic components is given by:

$$\sqrt{\sum_{n=2}^{\infty} V_{n,\text{RMS}}^2} = 0.055$$

THD is calculated as:

$$\text{THD} = \frac{\sqrt{\sum_{n=2}^{\infty} V_{n,\text{RMS}}^2}}{V_{f,\text{RMS}}}$$

Substituting the values:

$$\text{THD} = \frac{0.055}{2.699} = 0.020622$$

XXVI. Conclusion

This project successfully designed and implemented an audio amplifier with pre-amplification, gain, filtering, and power amplification stages. The system achieved high signal quality with minimal distortion, ensuring efficient and clear audio output. Simulations and hardware testing validated the design's stability and performance. Future improvements could enhance efficiency, filtering, and adaptive gain control for better audio fidelity.